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硕士学位论文

二维Helmholtz方程的高阶有限差分方法

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摘 要

Helmholtz方程是一类重要的椭圆型偏微分方程，在许多物理现象中，都涉及到Helmholtz方程，包括在时间调和的声音和电磁场中的势能，声波的散射，消声器中噪音的降低，水波的蔓延和膜的振动等。得到Helmholtz方程的有效和精确的数值解已经成为许多研究的主题。若干种求解Helmholtz方程数值解的方法已经发展起来，例如有限差分方法，边界元方法，有限元方法和谱元素方法。

有限差分方法是用于求解微分方程问题的最广泛的数值方法，很多常见的偏微分方程可用这种方法得到近似解。在这篇文章中，我们推广一种多步有限差分方法，来求解二维的Helmholtz方程。通过这种方法，我们得出二维Helmholtz方程新的高阶差分格式，并对于Neumann边界条件，给出一个新的六阶差分格式。同时，文章中给出了两个数值算例，验证了我们提出的新的差分格式。

关键词： *Helmholtz*方程；有限差分方法；多步方法

Abstract

Helmholtz equation is an important kind of elliptic partial differential equation, which is involved in many physical phenomena, including the potential in time harmonic and electromagnetic fields, acoustic wave scattering, noise reduction in silencers, water wave propagation and membrane vibration. Obtaining an efficient and accurate numerical solution for the Helmholtz equation has been the subject of many studies. Many numerical methods are developed to solve Helmholtz equation such as the finite difference method, the boundary element method, the finite element method and the spectral element method.

The finite difference method is widely used to solve partial differential equations. In this paper, we promote a kind of multistep finite difference method in one dimension, and use it to solve Helmholtz equation in two dimensions. By this method, we obtain new high-order schemes of Helmholtz equation and a six-order scheme for Neumann boundary condition. Numerical results are presented on two problems approximated with our new schemes.

Keywords: Helmholtz equation; Finite difference method; Multistep method

第一章 背景知识

1.1 引言

Helmholtz方程是一个椭圆型的偏微分方程，其形式为

$$\Delta u(\mathbf{r}) + k^2 u(\mathbf{r}) = f(\mathbf{r}), \quad \mathbf{r} \in \Omega,$$

其中， Δ 是Laplace算子，它的Dirichlet边界条件可写成

$$u(\mathbf{r}_b) = u_0(\mathbf{r}_b), \quad \mathbf{r}_b \in \partial\Omega.$$

在许多重要的物理现象中，都涉及到Helmholtz方程，包括在时间调和的声音和电磁场中的势能，声波的散射，消声器中噪音的降低，水波的蔓延和膜的振动等。得到Helmholtz方程的有效和精确的数值解已经成为许多研究的主题。若干种求解Helmholtz方程的数值解的方法已经发展起来，例如有限差分方法，边界元方法，有限元方法和谱元素方法。

很多长期以来用来求解Helmholtz方程的经典方法都是建立在边界元方法的基础上的。这些方法是基于利用所要求解的问题的边界上的积分的思想，参见参考文献[5],[6],[23]。但从数值观点来看，边界元方法的缺点是在计算的过程中，会得出要花费大量时间求解的病态的方程组，尤其是对于三维的问题。

有限元方法被大量用来求解Helmholtz方程，见参考文献[1],[5],[11],[12],[27]。但有限元方法求解Helmholtz方程的缺点，除了计算量比较大之外，还有当 k 增大时，解的精确度会降低。所以，为了得到同样的解的精确度，就需要更多的结点。

谱元素方法在求解Helmholtz方程时，跟有限元方法相比需要较少的节点。但谱元素方法导出的系数矩阵没有有限元方法得出的系数矩阵那么稀疏，所以两种方法总的计算时间基本一样，参见参考文献[19]。

有限差分方法是用于求解微分方程问题的最广泛的数值方法，很多常见的偏微分方程可用这种方法得到近似解，见参考文献[16],[21],[22],[26],[35],[30],[38]。其基本思想是：用离散的、只含有有限个未知量的差分方程去近似代替连续变量的微分方程和定解条件，并把相应的差分方程的解作为微分方程定解问题的

近似解。同时,有限差分方法和其它方法相结合,例如多重网格方法,外推法,还可以由低阶格式得到高阶的数值解,可见参考文献[8],[18],[28],[33],[34],[36]。

参考文献[20],[24],[25]给出了一维和二维Helmholtz方程的高阶差分格式,参考文献[29]给出了三维Helmholtz方程的差分格式,参考文献[9]通过利用广义的导数的定义和Padé逼近,得出对于很大的 k 更稳定的差分格式。

在参考文献[15]中,作者提出了一种多步的有限差分方法。对于 d 阶线性常微分方程

$$y^{(d)}(x) = f(x)y(x) + g(x), \quad x_L \leq x \leq x_R,$$

其中, $f(x)$ 和 $g(x)$ 是 $[x_L, x_R]$ 上的连续函数。定义算子

$$L[y(x); h] \equiv \sum_{j=0}^k [\alpha_j y(x+jh) - h^d \beta_j y^{(d)}(x+jh)],$$

其中, h 是把区间 $[x_L, x_R]$ 分成 N 个等长的子区间后的步长,即 $h = (x_R - x_L)/N$ 。

将 $y(x+jh)$ 和 $y^{(d)}(x+jh)$ 在 x 处Taylor展开,可得到

$$L[y(x); h] = C_0 y(x) + C_1 h y'(x) + \cdots + C_p h^p y^{(p)}(x) + \cdots.$$

比较 $y(x)$ 的各阶导数的系数,并选择合适的 k, α_j, β_j ,可使得 $C_0 = C_1 = \cdots = C_p = 0$,同时 $C_{p+1} \neq 0$ 。由此,得到

$$\sum_{j=0}^k [\alpha_j y(x+jh) - h^d \beta_j y^{(d)}(x+jh)] = L[y(x); h] = C_{p+1} h^{p+1} y^{(p+1)}(x) + O(h^{p+2}).$$

通过略去右端项,并用 $f(x+jh)y(x+jh) + g(x+jh)$ 替换 $y^{(d)}(x+jh)$,就可得到一个局部截断误差为 $C_{p+1} h^{p+1} y^{(p+1)}(x) + O(h^{p+2})$ 的有限差分格式。

在这篇文章中,我们推广上述方法,得到二维Helmholtz方程的高阶的有限差分格式。在本章的余下部分,我们将给出计算二维Helmholtz方程数值解常用的差分格式;在第二章中,我们将导出新的高阶差分格式,并对于Neumann边界条件,给出一个新的六阶差分格式;在第三章中,我们将用数值实验,对我们所提出的差分格式进行验证,并和已有的差分格式进行比较。

1.2 求解Helmholtz方程常用的数值方法

1.2.1 有限元方法

假设 $\Omega \subset R^n (n = 1, 2)$ 是一个开的有界区域,并且它的边界 Γ 是Lipschitz连续的,这保证了在 Γ 上几乎处处可定义的单位法向量 $\mathbf{n}$ 的存在性。考虑如下的边

界值问题:

$$\begin{aligned}
 -\Delta u - k^2 u &= f && \text{在 } \Omega \text{ 里,} \\
 u &= g && \text{在 } \Gamma_g \text{ 上,} \\
 \nabla u \cdot \mathbf{n} &= q && \text{在 } \Gamma_q \text{ 上,} \\
 \nabla u \cdot \mathbf{n} + \alpha u &= r && \text{在 } \Gamma_r \text{ 上,}
 \end{aligned} \tag{1.1}$$

其中, $\Gamma_g \cup \Gamma_q \cup \Gamma_r = \Gamma$, 并且 $\Gamma_g \cap \Gamma_q = \Gamma_g \cap \Gamma_r = \Gamma_q \cap \Gamma_r = \emptyset$.

引入

$$\begin{aligned}
 U &= \{u \in H^1(\Omega) : u = g \text{ 在 } \Gamma_g \text{ 上}\}, \\
 V &= \{v \in H^1(\Omega) : v = 0 \text{ 在 } \Gamma_g \text{ 上}\}.
 \end{aligned}$$

其中, $H^1(\Omega)$ 是标准的 Hilbert 空间, 边界值问题 (1.1) 可以用如下的弱形式重新表述:

求出 $u \in U$, 满足

$$a(u, v) = f(v) \quad \forall v \in V, \tag{1.2}$$

其中,

$$\begin{aligned}
 a(u, v) &= \int_{\Omega} [\nabla u \cdot \nabla v - k^2 uv] d\Omega + \int_{\Gamma_r} \alpha uv d\Gamma, \\
 f(v) &= \int_{\Gamma_q} qv d\Gamma + \int_{\Gamma_r} rv d\Gamma + \int_{\Omega} f v d\Omega.
 \end{aligned}$$

让我们假设 $M^h = \{\Omega_1, \dots, \Omega_{ne}\}$ 是 Ω 的一个由标准的 C^0 元素 (三角形或者四边形) 组成的有限元剖分, 它的边界为 Γ_e , 满足:

$$\bar{\Omega} = \bigcup_{e=1}^{ne} \bar{\Omega}_e, \bar{\Omega} = \Omega \cup \Gamma, \bar{\Omega}_e = \Omega_e \cup \Gamma_e,$$

$$\text{并且} \quad \Omega_e \cap \Omega_{e'} = \emptyset \quad \text{如果 } e \neq e',$$

同时, 假设有限元子集 $U^{h,l} \subset U$ 和子空间 $V^{h,l} \subset V$:

$$U^{h,l} = \{u^h \in H^1(\Omega) : u_h^e \in P^l(\Omega_e) \quad \text{并且} \quad u^h = g^h \text{ 在 } \Gamma_g \text{ 上}\},$$

$$V^{h,l} = \{v^h \in H^1(\Omega) : v_h^e \in P^l(\Omega_e) \quad \text{并且} \quad v^h = 0 \text{ 在 } \Gamma_g \text{ 上}\},$$

其中, h 是网格的步长, $P^l(\Omega_e)$ 是阶数为 l 的多项式空间, g^h 是 g 的插值, u_h^e 表示 u^h 在每一个元素 Ω_e 上的限制。

所以, 问题(1.2)的有限元逼近就是求出 $u^h \in U^{h,l}$, 满足

$$a(u^h, v^h) = f(v^h) \quad \forall v^h \in V^{h,l}.$$

然后, 我们引入惯用的全局有限元展开

$$u(x, y) = \sum_{j=1}^{N_G} u_j \phi_j(x, y), \quad (1.3)$$

其中, N_G 是全局结点 (x_j, y_j) 的个数, 包括 $N_1 = N_G - N_B$ 个区域内部的结点和 N_B 个边界上的结点, $u_j = u(x_j, y_j)$ 是 u 在结点的函数值。全局插值函数 ϕ_j , 关于第 j 个全局结点满足条件: $\phi_j(x_j, y_j) = \delta_{i,j}$, 其中, $\delta_{i,j}$ 是 Kronecker 符号。

把(1.3)代入(1.2), 并令 $v = \phi_i, i = 1 = \dots = N_G$, 得到

$$\sum_{j=1}^{N_G} L_{ij} u_j - k^2 \sum_{j=1}^{N_G} M_{ij} u_j + \alpha \sum_{j=1}^{N_G} K_{ij} u_j = \tilde{f}_i, \quad (1.4)$$

其中,

$$L_{ij} = \int_{\Omega} \nabla \phi_i \cdot \nabla \phi_j d\Omega, \quad M_{ij} = \int_{\Omega} \phi_i \phi_j d\Omega, \quad K_{ij} = \int_{\Gamma_r} \phi_i \phi_j d\Gamma,$$

$$\tilde{f}_i = \int_{\Gamma_q} \phi_i q d\Gamma + \int_{\Gamma_r} \phi_i r d\Gamma + \int_{\Omega} \phi_i f d\Omega.$$

将 q, r, f 进行线性元展开, 并对(1.4)的右端做周线积分, 可得到关于 $\mathbf{u} = (u_1, \dots, u_{N_G})^T$ 的线性方程组。解此方程组, 便可得到 u 关于 ϕ_i 的线性表示, 从而求出 u 在结点处的值。

参考文献[10]给出了有限元取线性元时的误差分析: 当 $kh < 1$ 时, 相对误差 $e_h \leq C_1 kh + C_2 k^3 h^2$, 其中, C_1, C_2 是与 k 和 h 无关的常数。另外, 参考文献[11]和[12]给出了用任意阶多项式做有限元求解一维 Helmholtz 方程时的误差结果。

1.2.2 边界元方法

考虑边界值问题(1.1), 令 $\Omega \subset R^2$, 且假设 $\Gamma_r = \emptyset, f \equiv 0$ 。

在Helmholtz方程两边同时乘以 u^* ，并在 Ω 上做积分，可得

$$\int_{\Omega} (\Delta u(\mathbf{x}) + k^2 u(\mathbf{x})) u^*(\mathbf{x}) d\Omega_{\mathbf{x}} = 0. \quad (1.5)$$

假设 u^* 足够光滑，对(1.5)使用Green定理，可得

$$\int_{\Omega} (\Delta u^*(\mathbf{x}) + k^2 u^*(\mathbf{x})) u d\Omega_{\mathbf{x}} = - \int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial \mathbf{n}} u^*(\mathbf{x}) d\Gamma_{\mathbf{x}} + \int_{\Gamma} \frac{\partial u^*(\mathbf{x})}{\partial \mathbf{n}} u(\mathbf{x}) d\Gamma_{\mathbf{x}}. \quad (1.6)$$

取 $u^*(\mathbf{x})$ 为基本解，即Helmholtz方程

$$\Delta u(\mathbf{x}) + k^2 u(\mathbf{x}) = -\delta(\mathbf{x} - \mathbf{y})$$

的解，则 $u^*(\mathbf{x}, \mathbf{y}) = \frac{i}{4} H_0^{(1)}(k|\mathbf{x} - \mathbf{y}|)$ ，其中， $H_0^{(1)}$ 为第一类Hankel函数。

代入(1.6)，可得

$$u(\mathbf{y}) = \int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial \mathbf{n}} u^*(\mathbf{x}, \mathbf{y}) d\Gamma_{\mathbf{x}} - \int_{\Gamma} \frac{\partial u^*(\mathbf{x}, \mathbf{y})}{\partial \mathbf{n}} u(\mathbf{x}) d\Gamma_{\mathbf{x}} \quad \mathbf{y} \in \Omega. \quad (1.7)$$

当 $\mathbf{y} \in \Gamma$ 时，假设 Γ_{ε} 是以 $\mathbf{y}$ 为中心的、半径为 ε 的半圆，且 Γ_{ε} 落在区域 Ω 的外部，而 Γ_{ε} 的两个端点在 Γ 上。通过计算，可得

$$\lim_{\varepsilon \rightarrow 0} \left(\int_{\Gamma_{\varepsilon}} \frac{\partial u(\mathbf{x})}{\partial \mathbf{n}} u^*(\mathbf{x}, \mathbf{y}) d\Gamma_{\mathbf{x}} - \int_{\Gamma_{\varepsilon}} \frac{\partial u^*(\mathbf{x}, \mathbf{y})}{\partial \mathbf{n}} u(\mathbf{x}) d\Gamma_{\mathbf{x}} \right) = -\frac{1}{2} u(\mathbf{y}) \quad \mathbf{y} \in \Gamma.$$

代入(1.7)，可得

$$\frac{1}{2} u(\mathbf{y}) = \int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial \mathbf{n}} u^*(\mathbf{x}, \mathbf{y}) d\Gamma_{\mathbf{x}} - \int_{\Gamma} \frac{\partial u^*(\mathbf{x}, \mathbf{y})}{\partial \mathbf{n}} u(\mathbf{x}) d\Gamma_{\mathbf{x}} \quad \mathbf{y} \in \Gamma. \quad (1.8)$$

在边界 Γ 上取 N 个结点 $\mathbf{x}_1, \dots, \mathbf{x}_N$ ，将 $\mathbf{x}_i$ 代入(1.8)，得到 N 个方程

$$\frac{1}{2} u(\mathbf{x}_i) = \int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial \mathbf{n}} u^*(\mathbf{x}, \mathbf{x}_i) d\Gamma_{\mathbf{x}} - \int_{\Gamma} \frac{\partial u^*(\mathbf{x}, \mathbf{x}_i)}{\partial \mathbf{n}} u(\mathbf{x}) d\Gamma_{\mathbf{x}}, \quad i = 1, \dots, N. \quad (1.9)$$

因为函数 u^* 是已知的，所以(1.9)中的 $u^*(\mathbf{x}, \mathbf{x}_i)$ 和 $\frac{\partial u^*(\mathbf{x}, \mathbf{x}_i)}{\partial \mathbf{n}}$ 是已知的。利用 $u(\mathbf{x}_i)$ 和 $\frac{\partial u(\mathbf{x}_i)}{\partial \mathbf{n}}$ ， $i = 1, \dots, N$ ，对 Γ 上的 u 和 $\frac{\partial u}{\partial \mathbf{n}}$ 做插值。假设有 N_1 个结点在 Γ_g 上，有 $N_2 = N - N_1$ 个结点在 Γ_q 上，则有 N_1 个 $\frac{\partial u(\mathbf{x}_i)}{\partial \mathbf{n}}$ 是未知的，有 N_2 个 $u(\mathbf{x}_i)$ 是未知的。利用方程组(1.9)，可求出这 N 个未知量。这样，对于 Γ 上的结点 $\mathbf{x}_i$ ，我们就得到 $u(\mathbf{x}_i)$ 和 $\frac{\partial u(\mathbf{x}_i)}{\partial \mathbf{n}}$ 的值，其中， $i = 1, \dots, N$ 。

假设 Ω 内部的 n 个节点为 $y_1, \dots, y_n$, 将 y_j 代入方程(1.7), 得到

$$u(y_j) = \int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial n} u^*(\mathbf{x}, y_j) d\Gamma_{\mathbf{x}} - \int_{\Gamma} \frac{\partial u^*(\mathbf{x}, y_j)}{\partial n} u(\mathbf{x}) d\Gamma_{\mathbf{x}}, \quad j = 1, \dots, n. \quad (1.10)$$

利用前面求出的 $u(\mathbf{x}_i)$ 和 $\frac{\partial u(\mathbf{x}_i)}{\partial n}$ 在 Γ 上做插值, 并利用已知的函数 u^* , 便可得到 $\int_{\Gamma} \frac{\partial u(\mathbf{x})}{\partial n} u^*(\mathbf{x}, y_j) d\Gamma_{\mathbf{x}}$ 和 $\int_{\Gamma} \frac{\partial u^*(\mathbf{x}, y_j)}{\partial n} u(\mathbf{x}) d\Gamma_{\mathbf{x}}$ 的值, 从而求出 $u(y_j)$ 。

1.3 二维Helmholtz方程常用的差分格式

1.3.1 内部点的差分格式

我们考虑实数值的二维Helmholtz方程:

$$\frac{\partial^2 u(x, y)}{\partial x^2} + \frac{\partial^2 u(x, y)}{\partial y^2} + k^2 u(x, y) = f(x, y), \quad (x, y) \in \Omega \quad (1.11)$$

我们假设 k 是常数。为了离散方程(1.11), 我们设区域 Ω 是长方形区域, 且用一致网格来离散 Ω 。设离散网格在 x 方向上和 y 方向上的步长都为 h 。

设 $u_{i,j}$ 是对 $u(x_i, y_j)$ 的数值逼近, 类似的, $f_{i,j} = f(x_i, y_j)$ 。我们希望导出在 x 方向和 y 方向上都对称的差分格式, 满足这种条件的差分格式的形式为

$$A_0 u_{i,j} + A_s \sigma_s + A_c \sigma_c = B_0 f_{i,j} + B_s \Gamma_s + B_c \Gamma_c, \quad (1.12)$$

其中,

$$\sigma_s = u_{i,j-1} + u_{i,j+1} + u_{i-1,j} + u_{i+1,j},$$

$$\sigma_c = u_{i-1,j-1} + u_{i+1,j-1} + u_{i-1,j+1} + u_{i+1,j+1}.$$

同样的, 对于 f , 有

$$\Gamma_s = f_{i,j-1} + f_{i,j+1} + f_{i-1,j} + f_{i+1,j},$$

$$\Gamma_c = f_{i-1,j-1} + f_{i+1,j-1} + f_{i-1,j+1} + f_{i+1,j+1}.$$

我们用 δ_x^1 来记 x 方向上的一阶中心差分, 即

$$\delta_x^1 u_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2h},$$

把 x 方向上的标准的二阶中心差分记为 δ_{xx}^2 , 即

$$\delta_{xx}^2 u_{i,j} = \frac{u_{i+1,j} + u_{i-1,j} - 2u_{i,j}}{h^2}.$$

对于充分光滑的函数 u , 我们有

$$\delta_{xx}^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{h^2}{12} \frac{\partial^4 u}{\partial x^4} + \frac{h^4}{360} \frac{\partial^6 u}{\partial x^6} + O(h^6).$$

对于 $\frac{\partial^2 u}{\partial y^2}$, 我们做同样的逼近, 并将两式相加, 得到

$$(\delta_{xx}^2 + \delta_{yy}^2)u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{h^2}{12} \left(\frac{\partial^4 u}{\partial x^4} + \frac{\partial^4 u}{\partial y^4} \right) + \frac{h^4}{360} \left(\frac{\partial^6 u}{\partial x^6} + \frac{\partial^6 u}{\partial y^6} \right) + O(h^6) \quad (1.13)$$

因此, 我们得到

$$(\delta_{xx}^2 + \delta_{yy}^2)u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + O(h^2).$$

代入Helmholtz方程中, 我们得到Helmholtz方程的一个二阶五点的差分格式

$$\begin{aligned} A_0 &= -4 + h^2 k^2, A_s = 1, A_c = 0, \\ B_0 &= h^2, B_s = B_c = 0. \end{aligned} \quad (1.14)$$

忽略(1.13)中的 $O(h^4)$ 项, 利用Helmholtz方程(1.11), 将 $\frac{\partial^4 u}{\partial x^4}$ 和 $\frac{\partial^4 u}{\partial y^4}$ 用 $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^2 u}{\partial y^2}$, $\frac{\partial^4 u}{\partial x^2 \partial y^2}$ 替换, 并对 $\frac{\partial^2 u}{\partial x^2}$, $\frac{\partial^2 u}{\partial y^2}$, $\frac{\partial^4 u}{\partial x^2 \partial y^2}$ 进行离散, 可得

$$(\delta_{xx}^2 + \delta_{yy}^2 + \frac{h^2}{6} \delta_{xx}^2 \delta_{yy}^2)u + (1 - \frac{h^2 k^2}{12})k^2 u = (1 - \frac{h^2 k^2}{12})f + \frac{h^2}{12}(\delta_{xx}^2 + \delta_{yy}^2)f + O(h^4). \quad (1.15)$$

利用

$$\delta_{xx}^2 \delta_{yy}^2 u = \frac{\sigma_c - 2\sigma_s + 4u_{i,j}}{h^4} + O(h^2),$$

我们得到Helmholtz方程的一个四阶的差分格式

$$\begin{aligned} A_0 &= -\frac{10}{3} + h^2 k^2 - \frac{h^4 k^4}{12}, A_s = \frac{2}{3}, A_c = \frac{1}{6}, \\ B_0 &= \frac{2h^2}{3} - \frac{h^2 k^2}{12}, B_s = \frac{h^2}{12}, B_c = 0. \end{aligned} \quad (1.16)$$

按照同样的思路, 利用Helmholtz方程(1.11), 把(1.13)中的 $\frac{\partial^6 u}{\partial x^6}, \frac{\partial^6 u}{\partial y^6}$ 用 u 的低阶导数替换, 再对低阶导数进行离散, 我们得到Helmholtz方程的一个六阶差分格式

$$\begin{aligned} A_0 u_{i,j} + A_s \sigma_s + A_c \sigma_c = & B_0 u_{i,j} + B_s \Gamma_s + B_c \Gamma_c + B_{11} \left(\frac{\partial^4 f_{i,j}}{\partial x^4} + \frac{\partial^4 f_{i,j}}{\partial y^4} \right) \\ & + B_{12} \frac{\partial^4 f_{i,j}}{\partial x^2 \partial y^2} + B_{13} \left(\frac{\partial^2 f_{i,j}}{\partial x^2} + \frac{\partial^2 f_{i,j}}{\partial y^2} \right), \end{aligned} \quad (1.17)$$

其中,

$$\begin{aligned} A_0 = & -\frac{10}{3} + \frac{46h^2 k^2}{45} - \frac{h^4 k^4}{12} + \frac{h^6 k^6}{360}, A_s = \frac{2}{3} - \frac{h^2 k^2}{90}, A_c = \frac{1}{6} + \frac{h^2 k^2}{180}, \\ B_0 = & h^2 - \frac{h^4 k^2}{12} + \frac{h^6 k^4}{360}, B_s = 0, B_c = 0, B_{11} = \frac{h^6}{360}, B_{12} = \frac{h^6}{90}, B_{13} = \frac{h^4}{12} - \frac{h^6 k^2}{360}. \end{aligned} \quad (1.18)$$

1.3.2 边界点的差分格式

二维Helmholtz方程的Neumann边界条件表示为

$$\frac{\partial u}{\partial n} = \theta(x, y).$$

当方程的定义域 Ω 为长方形区域时, Neumann边界条件可写成

$$\frac{\partial u}{\partial x} \Big|_{x=x_0} = \theta(y). \quad (1.19)$$

我们引入直线 $i = -1$, 最后我们将消去这条直线。由 $\delta_x^1 u_{i,j}$ 的Taylor展开, 得到

$$\delta_x^1 u_{i,j} = \frac{u_{i+1,j} - u_{i-1,j}}{2h} = \frac{\partial u_{i,j}}{\partial x} + \frac{h^2}{6} \frac{\partial^3 u_{i,j}}{\partial x^3} + O(h^4). \quad (1.20)$$

利用Helmholtz方程(1.11), 可以得到

$$\frac{\partial u_{i,j}}{\partial x} = \frac{u_{i+1,j} - u_{i-1,j}}{2h} + \frac{h^2}{6} \left(\frac{\partial^3 u_{i,j}}{\partial x \partial y^2} + k^2 \frac{\partial u_{i,j}}{\partial x} - \frac{\partial f_{i,j}}{\partial x} \right) + O(h^4). \quad (1.21)$$

再利用 $\frac{\partial u_{i,j}}{\partial x}$ 和 $\frac{\partial^3 u_{i,j}}{\partial x \partial y^2}$ 的二阶离散

$$\frac{\partial^3 u_{i,j}}{\partial x \partial y^2} = \delta_x^1 \delta_y^2 u_{i,j} + O(h^2),$$

和

$$\begin{aligned}\frac{\partial u_{i,j}}{\partial x} &= \frac{u_{i+1,j} - u_{i-1,j}}{2h} + O(h^2) = \frac{u_{i+1,j} - u_{i-1,j}}{2h} + \mu h^2 \frac{\partial^3 u}{\partial x \partial y^2} + O(h^2) \\ &= \frac{(1-2\mu)(u_{i+1,j} - u_{i-1,j})}{2h} + \mu \frac{u_{i+1,j+1} + u_{i+1,j-1} - u_{i-1,j+1} - u_{i-1,j-1}}{2h} + O(h^2),\end{aligned}$$

其中, μ 是任意参数。

代入(1.21), 两边乘以 $2h$, 并令 $i = 0$, 得到

$$\begin{aligned}2h\theta(y_j) &= u_{1,j} - u_{-1,j} + \frac{1}{6}[u_{1,j+1} - 2u_{1,j} + u_{1,j-1} - (u_{-1,j+1} - 2u_{-1,j} + u_{-1,j-1})] \\ &+ \frac{h^2 k^2}{6}[(1-2\mu)(u_{1,j} - u_{-1,j}) + \mu(u_{1,j+1} + u_{1,j-1} - u_{-1,j+1} - u_{-1,j-1})] - \frac{h^3}{3} \frac{\partial f_{0,j}}{\partial x}.\end{aligned}$$

这是一个四阶格式, 我们可以写成如下形式:

$$\hat{A}_s u_{1,j} + \hat{A}_c(u_{1,j+1} + u_{1,j-1}) - \hat{A}_s u_{-1,j} - \hat{A}_c(u_{-1,j+1} + u_{-1,j-1}) = 2h\theta(y_j) + \frac{h^3}{3} \frac{\partial f_{0,j}}{\partial x}. \quad (1.22)$$

我们注意到 μ 是任意的, 而且我们有

$$\hat{A}_s = \frac{2}{3} + \frac{h^2 k^2}{6}(1-2\mu), \hat{A}_c = \frac{1}{6} + \frac{h^2 k^2}{6}\mu. \quad (1.23)$$

在 $i = 0$ 这条线上, 内部点的标准逼近可以写成

$$\begin{aligned}A_s(u_{1,j} + u_{-1,j} + u_{0,j+1} + u_{0,j-1}) + A_c(u_{1,j+1} + u_{1,j-1} + u_{-1,j+1} + u_{-1,j-1}) + A_0 u_{0,j} \\ = B_0 f_{0,j} + B_s \Gamma_s + B_c \Gamma_c.\end{aligned} \quad (1.24)$$

我们希望利用方程(1.22)和(1.24)消去直线 $i = -1$ 上的元素。为了达到这个目的, 我们在方程(1.22)的两边乘以常数 α , 并和方程(1.24)相加。因此, α 应当满足

$$A_s = \alpha \hat{A}_s, A_c = \alpha \hat{A}_c,$$

即

$$\alpha = \frac{A_s}{\hat{A}_s} = \frac{A_c}{\hat{A}_c}.$$

我们解这个方程, 发现 α 存在解, 仅当以下条件满足

$$\frac{A_s}{\frac{2}{3} + \frac{h^2 k^2}{6}(1 - 2\mu)} = \frac{A_c}{\frac{1}{6} + \frac{h^2 k^2}{6}\mu},$$

即

$$\begin{aligned} \frac{\mu h^2 k^2}{6}(A_s + 2A_c) &= \frac{h^2 k^2}{6}A_c + \frac{2}{3}A_c - \frac{1}{6}A_s, \\ \mu h^2 k^2 &= \frac{(4 + h^2 k^2)A_c - A_s}{A_s + 2A_c}. \end{aligned}$$

我们选择 μ , 使其满足上述最后一个方程. 然后代入(1.23), 得到

$$\begin{aligned} \hat{A}_s &= \frac{2}{3} + \frac{h^2 k^2}{6} - \frac{1}{3} \frac{(4 + h^2 k^2)A_c - A_s}{A_s + 2A_c} = \left(1 + \frac{h^2 k^2}{6}\right) \frac{A_s}{A_s + 2A_c}, \\ \hat{A}_c &= \frac{1}{6} + \frac{1}{6} \frac{(4 + h^2 k^2)A_c - A_s}{A_s + 2A_c} = \left(1 + \frac{h^2 k^2}{6}\right) \frac{A_c}{A_s + 2A_c} = \frac{A_c}{A_s} \hat{A}_s, \\ \alpha &= \left(1 + \frac{h^2 k^2}{6}\right)^{-1} A_s + 2A_c. \end{aligned} \quad (1.25)$$

把(1.22), (1.24), (1.25)相加, 我们得到在 $i = 0$ 的一个四阶逼近格式

$$\begin{aligned} A_s u_{1,j} + A_c(u_{1,j+1} + u_{1,j-1}) + \frac{1}{2}A_s(u_{0,j-1} + u_{0,j+1}) + \frac{1}{2}A_0 u_{0,j} \\ = \frac{1}{2}B_0 f_{0,j} + \frac{1}{2}B_s \Gamma_s + \frac{1}{2}B_c \Gamma_c + \alpha(h\theta(y_j) + \frac{h^3}{6} \frac{\partial f_{0,j}}{\partial x}), \end{aligned} \quad (1.26)$$

其中, A 和 B 的值由内部点的四阶格式(1.16)给出。

第二章 二维Helmholtz方程的新的阶差分格式

2.1 内部点的高阶差分格式

在这一章中,我们将推广参考文献[15]中的多步有限差分方法,从而得出二维Helmholtz方程的高阶的差分格式。

定义算子

$$\begin{aligned} L_2[u(x, y); h] \equiv & \sum_{j=-1}^1 \sum_{i=-1}^1 \alpha_{i,j} u(x + ih, y + jh) \\ & - h^2 \sum_{j=-1}^1 \sum_{i=-1}^1 [\beta_{i,j} (\frac{\partial^2 u(x + ih, y + jh)}{\partial x^2} + \frac{\partial^2 u(x + ih, y + jh)}{\partial y^2})]. \end{aligned} \quad (2.1)$$

把 $u(x + ih, y + jh)$, $u_{xx}(x + ih, y + jh)$ 和 $u_{yy}(x + ih, y + jh)$ 在点 (x, y) 处展成Taylor级数, 得到

$$\begin{aligned} L_2[u(x, y); h] = & C_{00}u(x, y) \\ & + h[C_{10}\frac{\partial u(x, y)}{\partial x} + C_{01}\frac{\partial u(x, y)}{\partial y}] \\ & + h^2[C_{20}\frac{\partial^2 u(x, y)}{\partial x^2} + C_{11}\frac{\partial^2 u(x, y)}{\partial x \partial y} + C_{02}\frac{\partial^2 u(x, y)}{\partial y^2}] \\ & + \dots \\ & + h^p \sum_{i=0}^p C_{(p-i)i} \frac{\partial^p u(x, y)}{\partial x^{(p-i)} \partial y^i} \\ & + \dots \end{aligned} \quad (2.2)$$

其中, $C_{00}, C_{10}, C_{01}, \dots, C_{kl} \dots$ 为常数。

为了使数值解的精确度达到四阶, 则(2.2)中 $h^0, h^1, \dots, h^5$ 的系数应全为0, 即 $C_{00} = C_{10} = C_{01} = \dots = C_{50} = C_{41} = \dots = C_{05} = 0$ 。但此时共有21个方程, 而未知数 $\alpha_{i,j}, \beta_{i,j}$ 的个数只有18个。所以, 我们假设 $\alpha_{i,j} = \alpha_{j,i}, \beta_{i,j} = \beta_{j,i}$ 。在这个条件下, C_{ij} 满足 $C_{ij} = C_{ji}$, 则方程的个数减少到12个, 而未知数的个数也同样有12个: $\alpha_{0,0}, \alpha_{0,1}, \alpha_{0,-1}, \alpha_{1,1}, \alpha_{-1,1}, \alpha_{-1,-1}, \beta_{0,0}, \beta_{0,1}, \beta_{0,-1}, \beta_{1,1}, \beta_{-1,1}, \beta_{-1,-1}$ 。

通过比较(2.1)和(2.2), 可得到如下线性方程组:

$$\begin{aligned}
 C_{00} &= \sum_{j=-1}^1 \sum_{i=-1}^1 \alpha_{i,j} = 0, \\
 C_{10} &= \sum_{j=-1}^1 (-\alpha_{-1,j} + \alpha_{1,j}) = 0, \\
 C_{20} &= \frac{1}{2} \sum_{j=-1}^1 (\alpha_{-1,j} + \alpha_{1,j}) - \sum_{j=-1}^1 \sum_{i=-1}^1 \beta_{i,j} = 0, \\
 C_{11} &= \alpha_{-1,-1} + \alpha_{1,1} - \alpha_{1,-1} - \alpha_{-1,1} = 0, \\
 C_{30} &= \frac{1}{3!} \sum_{j=-1}^1 (-\alpha_{-1,j} + \alpha_{1,j}) - \sum_{j=-1}^1 (-\beta_{-1,j} + \beta_{1,j}) = 0, \\
 C_{21} &= \frac{1}{4!} \sum_{j=-1}^1 (\alpha_{-1,j} + \alpha_{1,j}) - \frac{1}{2} \sum_{j=-1}^1 (\beta_{-1,j} + \beta_{1,j}) = 0, \\
 C_{31} &= \frac{4}{4!} (\alpha_{-1,-1} + \alpha_{1,1} - \alpha_{1,-1} - \alpha_{-1,1}) - (\beta_{-1,-1} + \beta_{1,1} - \beta_{1,-1} - \beta_{-1,1}) = 0, \\
 C_{22} &= \frac{6}{4!} (\alpha_{-1,-1} + \alpha_{1,1} + \alpha_{1,-1} + \alpha_{-1,1}) - \frac{1}{2} \sum_{j=-1}^1 (\beta_{-1,j} + \beta_{1,j}) - \frac{1}{2} \sum_{i=-1}^1 (\beta_{i,-1} + \beta_{i,1}) = 0, \\
 C_{50} &= \frac{1}{5!} \sum_{j=-1}^1 (-\alpha_{-1,j} + \alpha_{1,j}) - \frac{1}{3!} \sum_{j=-1}^1 (-\beta_{-1,j} + \beta_{1,j}) = 0, \\
 C_{41} &= \frac{5}{5!} (\alpha_{-1,1} + \alpha_{1,1} - \alpha_{1,-1} - \alpha_{-1,-1}) - \frac{3}{3!} (\beta_{-1,1} + \beta_{1,1} - \beta_{1,-1} - \beta_{-1,-1}) = 0, \\
 C_{32} &= \frac{10}{5!} (\alpha_{-1,1} + \alpha_{1,1} - \alpha_{1,-1} - \alpha_{-1,-1}) - \frac{3}{3!} (\beta_{1,-1} + \beta_{1,1} - \beta_{-1,1} - \beta_{-1,-1}) \\
 &\quad - \frac{1}{3!} \sum_{j=-1}^1 (-\beta_{-1,j} + \beta_{1,j}) = 0.
 \end{aligned} \tag{2.3}$$

解此方程组, 得到 $\alpha_{i,j}, \beta_{i,j}$ 的一族解: $(\alpha_{0,0}, \alpha_{0,1}, \alpha_{1,0}, \alpha_{-1,0}, \alpha_{0,-1}, \alpha_{1,1}, \alpha_{-1,1}, \alpha_{1,-1}, \alpha_{-1,-1}) = (-20a, 4a, 4a, 4a, 4a, a, a, a, a), (\beta_{0,0}, \beta_{0,1}, \beta_{0,-1}, \beta_{1,0}, \beta_{-1,0}, \beta_{1,1}, \beta_{-1,1}, \beta_{1,-1}, \beta_{-1,-1}) = (4a+4b, \frac{a-4b}{2}, \frac{a-4b}{2}, \frac{a-4b}{2}, \frac{a-4b}{2}, b, b, b, b)$, 其中, a, b 为任意参数, 且 $a \neq 0$ 。

令 $a = 1, b = 0$, 我们得到二维的Helmholtz方程一个四阶的有限差分格式:

$$\begin{aligned}
 A_0 &= -\frac{10}{3} + \frac{2}{3}h^2k^2, A_s = \frac{2}{3} + \frac{1}{12}h^2k^2, A_c = \frac{1}{6}, \\
 B_0 &= \frac{2}{3}h^2, B_s = \frac{h^2}{12}, B_c = 0.
 \end{aligned} \tag{2.4}$$

然后, 我们推导出Helmholtz方程的一个六阶的有限差分格式。假设, $(\alpha_{0,0}, \alpha_{0,1}, \alpha_{1,0}, \alpha_{-1,0}, \alpha_{0,-1}, \alpha_{1,1}, \alpha_{-1,1}, \alpha_{1,-1}, \alpha_{-1,-1}) = (-20a, 4a, 4a, 4a, 4a, a, a, a, a)$, $(\beta_{0,0}, \beta_{0,1}, \beta_{0,-1}, \beta_{1,0}, \beta_{-1,0}, \beta_{1,1}, \beta_{-1,1}, \beta_{1,-1}, \beta_{-1,-1}) = (4a+4b, \frac{a-4b}{2}, \frac{a-4b}{2}, \frac{a-4b}{2}, \frac{a-4b}{2}, b, b, b, b)$, 此时, $C_{60} = C_{06} = -\frac{a}{40}$, $C_{51} = C_{15} = C_{33} = 0$, $C_{42} = C_{24} = \frac{a}{24} - b$, $C_{70} = C_{61} = C_{52} = C_{43} = C_{34} = C_{25} = C_{16} = C_{07} = 0$ 。

把 $\alpha_{i,j}, \beta_{i,j}, C_{ij}$ 的值带入(2.1)和(2.2), 我们得到

$$\begin{aligned} & -20au_{i,j} + 4a\sigma_s + a\sigma_c - h^2[(4a+4b)(\frac{\partial^2 u_{i,j}}{\partial x^2} + \frac{\partial^2 u_{i,j}}{\partial y^2}) + \frac{a-4b}{2}(\frac{\partial^2 u_{i-1,j}}{\partial x^2} + \frac{\partial^2 u_{i-1,j}}{\partial y^2}) \\ & + \frac{\partial^2 u_{i+1,j}}{\partial x^2} + \frac{\partial^2 u_{i+1,j}}{\partial y^2} + \frac{\partial^2 u_{i,j-1}}{\partial x^2} + \frac{\partial^2 u_{i,j-1}}{\partial y^2} + \frac{\partial^2 u_{i,j+1}}{\partial x^2} + \frac{\partial^2 u_{i,j+1}}{\partial y^2}) + b(\frac{\partial^2 u_{i-1,j-1}}{\partial x^2} \\ & + \frac{\partial^2 u_{i-1,j-1}}{\partial y^2} + \frac{\partial^2 u_{i+1,j-1}}{\partial x^2} + \frac{\partial^2 u_{i+1,j-1}}{\partial y^2} + \frac{\partial^2 u_{i-1,j+1}}{\partial x^2} + \frac{\partial^2 u_{i-1,j+1}}{\partial y^2} + \frac{\partial^2 u_{i+1,j+1}}{\partial x^2} + \\ & \frac{\partial^2 u_{i+1,j+1}}{\partial y^2})] = h^6[-\frac{a}{40}(\frac{\partial^6 u_{i,j}}{\partial x^6} + \frac{\partial^6 u_{i,j}}{\partial y^6}) + (\frac{a}{24} - b)(\frac{\partial^6 u_{i,j}}{\partial x^4 \partial y^2} + \frac{\partial^6 u_{i,j}}{\partial x^2 \partial y^4})] + O(h^8). \end{aligned} \quad (2.5)$$

(2.5)左边的 $\frac{\partial^2 u_{i,j}}{\partial x^2} + \frac{\partial^2 u_{i,j}}{\partial y^2}$ 可直接用方程(1.11)替换。对于(2.5)的右边(RHS), 我们利用方程(1.11), 把 $u(x,y)$ 的高阶导数用低阶导数替换, 再对低阶导数进行离散。

对方程(1.11)的两边分别做 x 的四节偏导和 y 的四节偏导, 得到

$$\begin{aligned} \frac{\partial^6 u}{\partial x^6} + \frac{\partial^6 u}{\partial x^4 \partial y^2} + k^2 \frac{\partial^4 u}{\partial x^4} &= \frac{\partial^4 f}{\partial x^4}, \\ \frac{\partial^6 u}{\partial y^6} + \frac{\partial^6 u}{\partial x^2 \partial y^4} + k^2 \frac{\partial^4 u}{\partial y^4} &= \frac{\partial^4 f}{\partial y^4}. \end{aligned} \quad (2.6)$$

把(2.6)代入RHS, 得到:

$$\begin{aligned} RHS &= h^6[-\frac{a}{40}(\frac{\partial^4 f_{i,j}}{\partial x^4} + \frac{\partial^4 f_{i,j}}{\partial y^4}) + \frac{a}{40}k^2(\frac{\partial^4 u_{i,j}}{\partial x^4} + \frac{\partial^4 u_{i,j}}{\partial y^4}) + (\frac{a}{15} - b)(\frac{\partial^6 u_{i,j}}{\partial x^4 \partial y^2} + \frac{\partial^6 u_{i,j}}{\partial x^2 \partial y^4})] \\ &+ O(h^8). \end{aligned}$$

对方程(1.11)的两边做 x 的二节偏导和 y 的二节偏导, 得到

$$\frac{\partial^6 u}{\partial x^4 \partial y^2} + \frac{\partial^6 u}{\partial x^2 \partial y^4} + k^2 \frac{\partial^4 u}{\partial x^2 \partial y^2} = \frac{\partial^4 f}{\partial x^2 \partial y^2}. \quad (2.7)$$

再对方程(1.11)的两边分别做 x 的二阶偏导和 y 的二阶偏导, 得到

$$\begin{aligned}\frac{\partial^4 u}{\partial x^4} + \frac{\partial^4 u}{\partial x^2 \partial y^2} + k^2 \frac{\partial^2 u}{\partial x^2} &= \frac{\partial^2 f}{\partial x^2}, \\ \frac{\partial^4 u}{\partial y^4} + \frac{\partial^4 u}{\partial x^2 \partial y^2} + k^2 \frac{\partial^2 u}{\partial y^2} &= \frac{\partial^2 f}{\partial y^2}.\end{aligned}\quad (2.8)$$

把(2.7)和(2.8)代入RHS, 得到:

$$\begin{aligned}RHS = h^6 \{ & -\frac{a}{40} (\frac{\partial^4 f_{i,j}}{\partial x^4} + \frac{\partial^4 f_{i,j}}{\partial y^4}) + \frac{a}{40} k^2 [\frac{\partial^2 f_{i,j}}{\partial x^2} + \frac{\partial^2 f_{i,j}}{\partial y^2} - k^2 (f_{i,j} - k^2 u_{i,j}) - 2 \frac{\partial^4 u_{i,j}}{\partial x^2 \partial y^2}] \\ & + (\frac{a}{15} - b) (\frac{\partial^4 f_{i,j}}{\partial x^2 \partial y^2} - k^2 \frac{\partial^4 u_{i,j}}{\partial x^2 \partial y^2}) \} + O(h^8).\end{aligned}$$

对 $\frac{\partial^4 u_{i,j}}{\partial x^2 \partial y^2}$, $\frac{\partial^4 f_{i,j}}{\partial x^2 \partial y^2}$ 进行二阶离散, 得到

$$\begin{aligned}\frac{\partial^4 u_{i,j}}{\partial x^2 \partial y^2} &= \delta_{xx}^2 \delta_{yy}^2 u_{i,j} + O(h^2), \\ \frac{\partial^4 f_{i,j}}{\partial x^2 \partial y^2} &= \delta_{xx}^2 \delta_{yy}^2 f_{i,j} + O(h^2).\end{aligned}$$

代入RHS, 可得到二维Helmholtz方程的一个六阶的有限差分格式:

$$\begin{aligned}A_0 &= -20 + \frac{67h^2k^2}{15} - \frac{h^6k^6}{40}, A_s = 4 + \frac{4h^2k^2}{15}, A_c = 1 + \frac{7h^2k^2}{60}, \\ B_0 &= 4h^2 - \frac{h^6k^4}{40}, B_s = \frac{h^2}{2}, B_c = 0, B_{11} = -\frac{h^6}{40}, B_{12} = \frac{h^6}{15}, B_{13} = \frac{h^6k^2}{40}.\end{aligned}\quad (2.9)$$

若 f 的导数是未知的, 则用 $\delta_{xx}^2 f_{i,j}$, $\delta_{yy}^2 f_{i,j}$, $\delta_{xx}^2 \delta_{yy}^2 f_{i,j}$ 来得到 $\frac{\partial^2 f_{i,j}}{\partial x^2}$, $\frac{\partial^2 f_{i,j}}{\partial y^2}$ 和 $\frac{\partial^4 f_{i,j}}{\partial x^2 \partial y^2}$ 的二阶离散. 用

$$\frac{\partial^4 f_{i,j}}{\partial x^4} = \frac{16(f_{i-1,j} + f_{i+1,j}) - 64(f_{i-\frac{1}{2},j} + f_{i+\frac{1}{2},j}) + 96f_{i,j}}{h^4} + O(h^2)$$

来得到 $\frac{\partial^4 f_{i,j}}{\partial x^4}$ 和 $\frac{\partial^4 f_{i,j}}{\partial y^4}$ 的二阶离散, 其中, $f_{i \pm \frac{1}{2},j} = f(x_i \pm \frac{h}{2}, y_j)$.

2.2 边界点的六阶差分格式

对于二维Helmholtz方程的数值逼近, 我们希望在边界上得出和内部点一样的精确度. 对于Neumann边界条件, 我们将在这一节得出一个六阶的有限差分格式.

为了得到 $\frac{\partial u_{0,j}}{\partial x}$ 的六阶离散, 我们需要利用 $u_{-1,j}, u_{-1,j-1}, u_{-1,j+1}, u_{0,j}, u_{0,j-1}, u_{0,j+1}, u_{1,j}, u_{1,j-1}, u_{1,j+1}, u_{2,j}, u_{3,j}, u_{4,j}$ 这十二个点, 其中, $u_{-1,j} = u(x_0 - h, y_j)$ 。再利用第二章得到的六阶差分格式, 令 $i = 0$, 消去 $u_{-1,j}, u_{-1,j-1}, u_{-1,j+1}$ 。

设

$$\begin{aligned} & \left(\sum_{i=-1}^1 \sum_{k=-1}^1 \alpha_{i,j+k} u_{i,j+k} + \alpha_{2,j} u_{2,j} + \alpha_{3,j} u_{3,j} + \alpha_{4,j} u_{4,j} \right) - h^2 \left[\sum_{i=-1}^1 \sum_{k=-1}^1 \right. \\ & \quad \beta_{i,j+k} \left(\frac{\partial^2 u_{i,j+k}}{\partial x^2} + \frac{\partial^2 u_{i,j+k}}{\partial y^2} \right) + \beta_{2,j} \left(\frac{\partial^2 u_{2,j}}{\partial x^2} + \frac{\partial^2 u_{2,j}}{\partial y^2} \right) + \beta_{3,j} \left(\frac{\partial^2 u_{3,j}}{\partial x^2} + \frac{\partial^2 u_{3,j}}{\partial y^2} \right) \\ & \quad \left. + \beta_{4,j} \left(\frac{\partial^2 u_{4,j}}{\partial x^2} + \frac{\partial^2 u_{4,j}}{\partial y^2} \right) \right] = C_{00} u_{0,j} + h \left(C_{10} \frac{\partial u_{0,j}}{\partial x} + C_{01} \frac{\partial u_{0,j}}{\partial y} \right) + h^2 \left(C_{20} \frac{\partial^2 u_{0,j}}{\partial x^2} + \right. \\ & \quad \left. C_{11} \frac{\partial^2 u_{0,j}}{\partial x \partial y} + C_{02} \frac{\partial^2 u_{0,j}}{\partial y^2} \right) + \cdots + h^6 \left(C_{60} \frac{\partial^6 u_{0,j}}{\partial x^6} + C_{51} \frac{\partial^6 u_{0,j}}{\partial x^5 \partial y} + \cdots + C_{06} \frac{\partial^6 u_{0,j}}{\partial y^6} \right) \\ & \quad + O(h^7). \end{aligned} \quad (2.10)$$

其中, $\frac{\partial u_{0,j}}{\partial x} = \theta(y_j)$ 。

为了得到Neumann边界条件的六阶差分格式, $C_{i,j}$ 需要满足

$$C_{10} = 1, C_{00} = C_{01} = C_{20} = C_{11} = \cdots = C_{60} = C_{51} = \cdots = C_{06} = 0. \quad (2.11)$$

再比较方程(2.10)的两边, 可求出 $\alpha_{i,j}, \beta_{i,j}$ 的值。同时, 为了能和方程(2.9)相减, 消去 $u_{-1,j}, u_{-1,j-1}, u_{-1,j+1}$, 还要满足 $\alpha_{-1,j} : \alpha_{-1,j-1} : \alpha_{-1,j+1} : \beta_{-1,j} : \beta_{-1,j-1} : \beta_{-1,j+1} = 4 : 1 : 1 : \frac{4}{15} : \frac{7}{60} : \frac{7}{60}$ 。

经计算发现, 不存在 $\alpha_{i,j}, \beta_{i,j}$, 满足上述条件(2.11)。所以我们放宽条件, 改为

$$\begin{aligned} & C_{00} = C_{01} = \cdots = C_{04} = C_{41} = C_{23} = C_{05} = C_{60} = C_{51} = \cdots = C_{06} = 0, \\ & C_{10} = 1, C_{50} = C_{14} = \frac{1}{2} C_{32}. \end{aligned} \quad (2.12)$$

在条件(2.12)的情形下, 方程(2.10)右边存在 $\frac{\partial^5 u_{0,j}}{\partial x^5}, \frac{\partial^5 u_{0,j}}{\partial x^3 \partial y^2}, \frac{\partial^5 u_{0,j}}{\partial x \partial y^4}$ 。利用方程(1.11), 用 $u_{0,j}$ 的低阶导数替换上述三项。

通过计算, 可求出满足条件(2.12)的 $\alpha_{i,j}, \beta_{i,j}$ 的值为 $(\alpha_{-1,j}, \alpha_{-1,j-1}, \alpha_{-1,j+1}, \alpha_{0,j}, \alpha_{0,j-1}, \alpha_{0,j+1}, \alpha_{1,j}, \alpha_{1,j-1}, \alpha_{1,j+1}, \alpha_{2,j}, \alpha_{3,j}, \alpha_{4,j}) = (\frac{4}{5}, \frac{1}{5}, \frac{1}{5}, -\frac{85}{12}, \frac{4}{5}, \frac{4}{5}, -1, -1, -3,$

$$\left(\frac{8}{15}, -\frac{1}{20}\right), (\beta_{-1,j}, \beta_{-1,j-1}, \beta_{-1,j+1}, \beta_{0,j}, \beta_{0,j-1}, \beta_{0,j+1}, \beta_{1,j}, \beta_{1,j-1}, \beta_{1,j+1}, \beta_{2,j}, \beta_{3,j}, \beta_{4,j}) = \left(\frac{4}{75}, \frac{7}{300}, \frac{7}{300}, \frac{35}{36}, \frac{2}{45}, \frac{2}{45}, -\frac{229}{225}, -\frac{61}{900}, -\frac{61}{900}, -\frac{8}{75}, -\frac{29}{225}, -\frac{3}{100}\right).$$

此时, 方程(2.10)的右边(rhs)为

$$rhs = h\theta(y_j) - \frac{2h^5}{225} \frac{\partial^5 u_{0,j}}{\partial x^5} - \frac{4h^5}{225} \frac{\partial^5 u_{0,j}}{\partial x^3 \partial y^2} - \frac{2h^5}{225} \frac{\partial^5 u_{0,j}}{\partial x \partial y^4} + O(h^7).$$

由(1.11)得到

$$\begin{aligned} \frac{\partial^5 u}{\partial x^5} + \frac{\partial^5 u}{\partial x^3 \partial y^2} &= \frac{\partial^3 f}{\partial x^3} - k^2 \frac{\partial^3 u}{\partial x^3}, \\ \frac{\partial^5 u}{\partial x^3 \partial y^2} + \frac{\partial^5 u}{\partial x \partial y^4} &= \frac{\partial^3 f}{\partial x \partial y^2} - k^2 \frac{\partial^3 u}{\partial x \partial y^2}. \end{aligned} \quad (2.13)$$

把(2.13)代入rhs, 得到

$$rhs = h\theta(y_j) - \frac{2h^5}{225} \left(\frac{\partial^3 f_{0,j}}{\partial x^3} - k^2 \frac{\partial^3 u_{0,j}}{\partial x^3} \right) - \frac{2h^5}{225} \left(\frac{\partial^3 f_{0,j}}{\partial x \partial y^2} - k^2 \frac{\partial^3 u_{0,j}}{\partial x \partial y^2} \right) + O(h^7).$$

再由(1.11)得到

$$\frac{\partial^3 u}{\partial x^3} + \frac{\partial^3 u}{\partial x \partial y^2} = \frac{\partial f}{\partial x} - k^2 \frac{\partial u}{\partial x}. \quad (2.14)$$

把(2.14)代入rhs, 得到

$$rhs = h\theta(y_j) - \frac{2h^5}{225} \left(\frac{\partial^3 f_{0,j}}{\partial x^3} + \frac{\partial^3 f_{0,j}}{\partial x \partial y^2} \right) + \frac{2h^5 k^2}{225} \frac{\partial f_{0,j}}{\partial x} - \frac{2h^5 k^2}{225} \frac{\partial u_{0,j}}{\partial x} + O(h^7).$$

对于 $-\frac{2h^5 k^2}{225} \frac{\partial u_{0,j}}{\partial x}$, 我们用 $u_{0,j}, u_{1,j}, u_{2,j}$ 得到一个二阶离散, 可得

$$\frac{\partial u_{0,j}}{\partial x} = \frac{4u_{1,j} - u_{2,j} - 3u_{0,j}}{2h} + O(h^2). \quad (2.15)$$

将(2.15)代入rhs, 整理(2.10), 得到

$$\begin{aligned} &\left(\frac{1}{5} + \frac{7h^2 k^2}{300}\right)(u_{-1,j-1} + u_{-1,j+1}) + \left(\frac{4}{5} + \frac{4h^2 k^2}{75}\right)u_{-1,j} + \left(-\frac{85}{12} + \frac{35h^2 k^2}{36} - \frac{3h^4 k^4}{225}\right)u_{0,j} \\ &+ \left(\frac{4}{5} + \frac{2h^2 k^2}{45}\right)(u_{0,j-1} + u_{0,j+1}) + \left(\frac{44}{5} - \frac{229h^2 k^2}{225} + \frac{4h^4 k^4}{225}\right)u_{1,j} + \left(-1 - \frac{61h^2 k^2}{900}\right)(u_{1,j-1} \\ &+ u_{1,j+1}) + \left(-3 - \frac{8h^2 k^2}{75} - \frac{h^4 k^4}{225}\right)u_{2,j} + \left(\frac{8}{15} + \frac{29h^2 k^2}{225}\right)u_{3,j} + \left(-\frac{1}{20} - \frac{3h^2 k^2}{100}\right)u_{4,j} \end{aligned}$$

$$\begin{aligned}
&= h\theta(y_j) - \frac{2h^5}{225} \left(\frac{\partial^3 f_{0,j}}{\partial x^3} + \frac{\partial^3 f_{0,j}}{\partial x \partial y^2} \right) + \frac{2h^5 k^2}{225} \frac{\partial f_{0,j}}{\partial x} + \frac{7h^2}{300} (f_{-1,j-1} + f_{-1,j+1}) + \frac{4h^2}{75} f_{-1,j} \\
&\quad + \frac{35h^2}{36} f_{0,j} + \frac{2h^2}{45} (f_{0,j-1} + f_{0,j+1}) - \frac{229h^2}{225} f_{1,j} - \frac{61h^2}{900} (f_{1,j-1} + f_{1,j+1}) - \frac{8h^2}{75} f_{2,j} \\
&\quad + \frac{29h^2}{225} f_{3,j} - \frac{3h^2}{100} f_{4,j} + O(h^7).
\end{aligned} \tag{2.16}$$

(2.16)与(2.9)相减, 便可消去 $u_{-1,j}$ 、 $u_{-1,j-1}$ 、 $u_{-1,j+1}$ 。对于(2.16)右边的 $f_{-1,j}$ 、 $f_{-1,j-1}$ 、 $f_{-1,j+1}$, 在 (x_0, y_j) 点做Taylor展开, 得到

$$\begin{aligned}
&\frac{7h^2}{300} (f_{-1,j-1} + f_{-1,j+1}) + \frac{4h^2}{75} f_{-1,j} = \frac{h^2}{10} f_{0,j} - \frac{h^3}{10} \frac{\partial f_{0,j}}{\partial x} + \frac{h^4}{20} \frac{\partial^2 f_{0,j}}{\partial x^2} - \frac{h^5}{60} \frac{\partial^3 f_{0,j}}{\partial x^3} + \\
&\quad \frac{h^6}{240} \frac{\partial^4 f_{0,j}}{\partial x^4} + h^4 \frac{\partial^2 f_{0,j}}{\partial y^2} - h^5 \frac{\partial^3 f_{0,j}}{\partial x \partial y^2} + \frac{h^6}{2} \frac{\partial^4 f_{0,j}}{\partial x^2 \partial y^2} + \frac{h^6}{12} \frac{\partial^4 f_{0,j}}{\partial y^4} + O(h^7).
\end{aligned} \tag{2.17}$$

最终, 我们得到Neumann边界条件的一个六阶差分格式

$$\begin{aligned}
&A_{00}u_{0,j} + A_{01}(u_{0,j-1} + u_{0,j+1}) + A_{10}u_{1,j} + A_{11}(u_{1,j-1} + u_{1,j+1}) + A_{20}u_{2,j} + A_{30}u_{3,j} \\
&\quad + A_{40}u_{4,j} = h\theta(y_j) + B_{00}f_{0,j} + B_{01}(f_{0,j-1} + f_{0,j+1}) + B_{10}f_{1,j} + B_{11}(f_{1,j-1} + f_{1,j+1}) \\
&\quad + B_{20}f_{2,j} + B_{30}f_{3,j} + B_{40}f_{4,j} + d_{10} \frac{\partial f_{0,j}}{\partial x} + d_{20} \frac{\partial^2 f_{0,j}}{\partial x^2} + d_{02} \frac{\partial^2 f_{0,j}}{\partial y^2} + d_{30} \frac{\partial^3 f_{0,j}}{\partial x^3} + \\
&\quad d_{12} \frac{\partial^3 f_{0,j}}{\partial x \partial y^2} + d_{40} \frac{\partial^4 f_{0,j}}{\partial x^4} + d_{22} \frac{\partial^4 f_{0,j}}{\partial x^2 \partial y^2} + d_{04} \frac{\partial^4 f_{0,j}}{\partial y^4},
\end{aligned} \tag{2.18}$$

其中,

$$\begin{aligned}
A_{00} &= -\frac{37}{12} + \frac{71h^2k^2}{900} - \frac{h^4k^4}{75} + \frac{h^6k^6}{200}, A_{01} = -\frac{2h^2k^2}{225}, A_{10} = 8 - \frac{241h^2k^2}{225} + \frac{4h^2k^2}{225}, \\
A_{11} &= -\frac{6}{5} - \frac{41h^2k^2}{450}, A_{20} = -3 - \frac{8h^2k^2}{75} - \frac{h^4k^4}{225}, A_{30} = \frac{8}{15} + \frac{29h^2k^2}{225}, \\
A_{40} &= -\frac{1}{20} - \frac{3h^2k^2}{100}, \\
B_{00} &= \frac{49h^2}{180} + \frac{h^6k^4}{200}, B_{01} = -\frac{h^2}{18}, B_{10} = -\frac{503h^2}{450}, B_{11} = -\frac{61h^2}{900}, B_{20} = -\frac{8h^2}{75}, \\
B_{30} &= \frac{29h^2}{225}, B_{40} = -\frac{3h^2}{100},
\end{aligned}$$

$$\begin{aligned}
d_{10} &= -\frac{h^3}{10} + \frac{2h^5k^2}{225}, \\
d_{20} &= \frac{h^4}{20} - \frac{h^6k^2}{200}, d_{02} = h^4 - \frac{h^6k^2}{200}, d_{30} = -\frac{23h^5}{900}, \\
d_{12} &= -\frac{227h^5}{225}, d_{40} = \frac{11h^6}{1200}, d_{22} = \frac{73h^6}{150}, d_{04} = \frac{53h^6}{600}.
\end{aligned} \tag{2.19}$$

若 f 的导数是未知的, 则用

$$\frac{\partial f_{0,j}}{\partial x} = \frac{-25f_{0,j} + 48f_{1,j} - 36f_{2,j} + 16f_{3,j} - 3f_{4,j}}{4h} + O(h^4)$$

来得到 $\frac{\partial f_{0,j}}{\partial x}$ 的四阶离散。用

$$\frac{\partial^2 f_{0,j}}{\partial x^2} = \frac{35f_{0,j} - 104f_{1,j} + 114f_{2,j} - 56f_{3,j} + 11f_{4,j}}{12h^2} + O(h^3)$$

来得到 $\frac{\partial^2 f_{0,j}}{\partial x^2}$ 的三阶离散。用

$$\frac{\partial^2 f_{0,j}}{\partial y^2} = \frac{-(f_{0,j-1} + f_{0,j+1}) + 8(f_{0,j-\frac{1}{2}} + f_{0,j+\frac{1}{2}}) - 14f_{0,j}}{h^2} + O(h^3)$$

来得到 $\frac{\partial^2 f_{0,j}}{\partial y^2}$ 的三阶离散。用

$$\frac{\partial^3 f_{0,j}}{\partial x^3} = \frac{-5f_{0,j} + 18f_{1,j} - 24f_{2,j} + 16f_{3,j} - 3f_{4,j}}{2h^3} + O(h^2)$$

来得到 $\frac{\partial^3 f_{0,j}}{\partial x^3}$ 的二阶离散。用

$$\begin{aligned}
\frac{\partial^3 f_{0,j}}{\partial x \partial y^2} &= [-(f_{1,j-1} + f_{1,j+1}) + 2f_{i,j} + 4(f_{\frac{1}{2},j-1} + f_{\frac{1}{2},j+1}) - 8f_{\frac{1}{2},j} - 3(f_{0,j-1} + f_{0,j+1}) + 6f_{0,j}]/h^3 \\
&\quad + O(h^2)
\end{aligned}$$

来得到 $\frac{\partial^3 f_{0,j}}{\partial x \partial y^2}$ 的二阶离散。用

$$\frac{\partial^4 f_{0,j}}{\partial x^4} = \frac{f_{0,j} - 4f_{1,j} + 6f_{2,j} - 4f_{3,j} + f_{4,j}}{h^4} + O(h)$$

来得到 $\frac{\partial^4 f_{0,j}}{\partial x^4}$ 的一阶离散。用

$$\frac{\partial^4 f_{0,j}}{\partial y^4} = \frac{48[(f_{0,j-1} + f_{0,j+1}) - 4(f_{0,j-\frac{1}{2}} + f_{0,j+\frac{1}{2}}) + 6f_{0,j}]}{h^4} + O(h)$$

来得到 $\frac{\partial^4 f_{0,j}}{\partial y^4}$ 的一阶离散。用

$$\begin{aligned}
\frac{\partial^4 f_{0,j}}{\partial x^2 \partial y^2} &= [4(f_{1,j-1} + f_{1,j+1}) - 8f_{i,j} - 8(f_{\frac{1}{2},j-1} + f_{\frac{1}{2},j+1}) + 16f_{\frac{1}{2},j} + 4(f_{0,j-1} + f_{0,j+1}) - 4f_{0,j}]/h^4 \\
&\quad + O(h)
\end{aligned}$$

来得到 $\frac{\partial^4 f_{0,j}}{\partial x^2 \partial y^2}$ 的一阶离散。

第三章 数值实验

3.1 数值算例

在这一章中，我们用数值算例验证第二章中得出的有限差分格式达到了六阶，并和已有的高阶差分格式进行比较。

对于差分格式的误差 e_h ，我们定义 l_∞ 范数为

$$\|e_h\|_\infty = \max_{i,j} |u(x_i, y_j) - u_{i,j}|,$$

其中， $u(x_i, y_j)$ 为解析解在 (x_i, y_j) 点的值， $u_{i,j}$ 为差分方法在 (x_i, y_j) 点得到的近似值。定义 l_2 范数为

$$\|e_h\|_2 = \frac{1}{N} \sqrt{\sum_{i,j} (u(x_i, y_j) - u_{i,j})^2},$$

其中， N 为节点数。

在估计差分格式的阶数的时候，对于步长 h ，我们设

$$\|e_h\|_\infty \approx Ch^m,$$

其中， m 是差分格式的阶数。令 $H = 2h$ ，则

$$\frac{\|e_H\|_\infty}{\|e_h\|_\infty} \approx \frac{CH^m}{Ch^m} = 2^m,$$

所以，

$$m \approx \log_2 \left(\frac{\|e_H\|_\infty}{\|e_h\|_\infty} \right).$$

算例1.对于第一章中提到的已有的差分格式(1.18)，我们称之为SFD；对于第二章我们提出的新的差分格式(2.9)，我们称之为NFD。在算例1中，我们将对两种差分格式进行比较。

我们要求解的Helmholtz方程为

$$\frac{\partial^2 u(x, y)}{\partial x^2} + \frac{\partial^2 u(x, y)}{\partial y^2} + k^2 u(x, y) = (k^2 - 32\pi^2) \sin(4\pi x) \sin(4\pi y), \quad (3.1)$$
$$(x, y) \in \Omega = [0, 1] \times [0, 1],$$

其中, 边界条件为

$$u(0, y) = u(1, y) = u(x, 0) = u(x, 1) = 0.$$

方程(3.1)的解析解为

$$u(x, y) = \sin(4\pi x) \sin(4\pi y).$$

我们令 $k = 10$ 。表格1显示出算例1的比较结果。

h	$\ e_h\ _2(\text{SFD})$	$\ e_h\ _\infty(\text{SFD})$	m(SFD)	$\ e_h\ _2(\text{NFD})$	$\ e_h\ _\infty(\text{NFD})$	m(NFD)
$\frac{1}{8}$	0.0023	0.0284	6.5246	8.7572e-004	0.0107	6.2285
$\frac{1}{16}$	1.0954e-005	3.0808e-004	6.1267	1.0954e-005	1.4307e-004	6.0503
$\frac{1}{32}$	7.3409e-008	4.4091e-006	6.0314	3.5943e-008	2.1588e-006	6.0121
$\frac{1}{64}$	5.4349e-010	6.7409e-008	6.0078	5.4349e-010	3.3450e-008	6.0030
$\frac{1}{128}$	4.1568e-012	1.0476e-009	5.9960	2.0695e-012	5.2157e-010	6.0223
$\frac{1}{256}$	3.2210e-014	1.6414e-011		1.5769e-014	8.0245e-012	

表格1

算例2. 在算例2中, 我们将验证差分格式(2.19)的精确度。

我们要求解的方程为

$$\begin{aligned} \frac{\partial^2 u(x, y)}{\partial x^2} + \frac{\partial^2 u(x, y)}{\partial y^2} + k^2 u(x, y) &= (k^2 - 2\pi^2) \sin(\pi x) \sin(\pi y), \\ (x, y) \in \Omega &= [0, 1] \times [0, 1], \end{aligned} \quad (3.2)$$

其中, 边界条件为

$$u(1, y) = u(x, 0) = u(x, 1) = 0, \quad \frac{\partial u}{\partial x} \Big|_{x=x_0} = \pi \sin(\pi y).$$

方程(3.2)的解为

$$u(x, y) = \sin(\pi x) \sin(\pi y).$$

我们称差分格式(1.26)为N-4, 称差分格式(2.19)为N-6. 令 $k = 10$. 表格2表明了两种差分格式的数值比较结果.

h	$\ e_h\ _2(N-6)$	$m(N-6)$	$\ e_h\ _2(N-4)$	$m(N-4)$
$\frac{1}{8}$	2.8248e-004	5.9508	0.0095	5.8648
$\frac{1}{16}$	4.5669e-006	5.9903	1.6302e-004	5.1346
$\frac{1}{32}$	7.1840e-008	5.9974	4.6406e-006	5.0302
$\frac{1}{64}$	1.1245e-009	6.0088	1.4201e-007	4.4425
$\frac{1}{128}$	1.7464e-011		6.5312e-009	

表格2

3.2 小结

在这篇文章中, 我们推广了参考文献[15]中的多步差分方法, 导出了二维Helmholtz方程的高阶差分格式, 并给出了Neumann边界条件的一个六阶格式. 本章中的数值例子验证了这两个格式的精确度都达到了六阶, 而且相比于已有的差分格式(1.18), 新的差分格式的误差要更小.

对于进一步的工作, 可以把这种多步方法用于其它二维偏微分方程, 或者推广到三维的方程上.

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